Hasan Forghani Ramandy

Full-Stack Developer (Web & Windows)

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M.Sc. in Physics (Computational Cosmology) with 7+ years of experience in programming and software development across multiple languages and frameworks. Strong foundation in algorithms and problem-solving, applied in large-scale web applications, CRMs, and desktop solutions. Passionate about learning new technologies and delivering high-quality, scalable solutions.

EXPERIENCES

Senior Web Developer — *Boboko Holding*, *Tehran, Iran*
As the Head of Web Development, I led a team specializing in the development and maintenance of robust online platforms. My expertise was primarily in back-end solutions with PHP, WordPress, and WooCommerce, where I focused on creating custom plugins and themes to meet unique business needs. I also provided key support for front-end development using JQuery, Bootstrap, and Elementor, ensuring an optimal user experience and strong site performance.

Senior Web Developer — *LoyalAxis*, *Tehran, Iran*
As Technical Manager, I oversaw the development of LoyalAxis CRM, a WordPress-based solution for small and medium-sized businesses. I designed and implemented the core application and over 10 modules, including advanced customer management, workflow automation, WooCommerce integration, and real-time analytics. This project demanded expertise in PHP, JavaScript, SQL, and modern practices (SOLID, TDD/BDD, Agile) along with technologies such as React, Postman, Docker, and Git to ensure scalability and high performance.

Windows App Developer — *Lavin24*, *Tehran, Iran*
As Full-Stack Developer, I developed Lavin24, a software for video encryption and customer licensing to protect commercial videos from unauthorized copying and distribution. The system consists of two Windows applications built with .NET (C#) using a complex RAM-buffer video encryption and playback algorithm, and a web solution based on WordPress for user management and license validation. This infrastructure is used by 500+ educational institutions and thousands of students, ensuring secure content delivery at scale.

Scientific Programmer — *University of Tehran*, *Tehran, Iran*
As the sole programmer in the Gravity and Cosmology Group at the University of Tehran, I developed a Python-based simulation to model light trajectories in metamaterials that mimic the optical behaviour of black holes. The program uses numerical methods to solve complex differential equations and visualizes results with Matplotlib. The underlying theory and results were published by American Physical Society (APS).

Web Developer — *Freelance*, *Tehran, Iran*
As a Freelance Web Developer, I delivered 20+ websites and web applications, focusing on performance, scalability, and user experience. I consulted startups on technology stack selection and system architecture, helping reduce time-to-market and improve maintainability. Projects required expertise PHP, Laravel, WordPress, WooCommerce, MySQL, JavaScript, HTML, CSS, and Linux, alongside Git-based workflows and automated deployment practices. I also gained experience in project management, client communication, and delivering solutions on time and within budget.

EDUCATION

University of Tehran
M.Sc. in Physics
Tehran, IRAN
Sept. 2017 – Sept. 2020

Sharif University of Technology
B.Sc. in Physics
Tehran, IRAN
Sept. 2011 – Sept. 2016

SKILLS

Programming Languages:

PHP	●●●●●
Python	●●●●●
C#	●●●●○
SQL	●●●○○
JavaScript	●●●○○

Languages Proficiency:

- English — *Fluent (Toefl:102)*
- Farsi — *Native*
- Arabic — *Elementary*

Backend Development:
Laravel, WordPress, Composer, WooCommerce, MySQL, Redis, Blade, SOLID, REST API/GraphQL, JWT/OAuth2, Microservices architecture, Domain-Driven Design (DDD), CSRF, XSS

Frontend Development:
React, Vue.js, jQuery, Elementor, Alpine.js, Tailwind, Bootstrap, Livewire, Inertia.js, Webpack, Vite, Sass, PostCSS, Accessibility (a11y)

Systems & Workflow:
Git, Docker, Linux, CI/CD, Caching strategies , Role-Based Access Control (RBAC)

PUBLICATIONS

Theses

- Forghani Ramandy, Hasan (Sept. 2020). "A study on metamaterial analogs of some of exact solutions of Einstein's field equations". Supervisor: Professor Mohammad Nouri Zonoz. MA thesis. University of Tehran. URL: https://drive.google.com/file/d/17yS06ks0rhfz5WtqaIkiuW2KOqvARps-/view?usp=sharing.

Articles

- Nouri-Zonoz, M., A. Parvizi, and H. Forghani-Ramandy (Dec. 2022). "Metamaterial analog of a black hole shadow: An exact ray-tracing simulation based on the spacetime index of refraction". In: *Phys. Rev. D* 106 (12), p. 124013. DOI: 10.1103/PhysRevD.106.124013. URL: https://link.aps.org/doi/10.1103/PhysRevD.106.124013.
- Parvizi, A., H. Forghani-Ramandy, and M. Nouri-Zonoz (Nov. 2024). "Photon rings in the metamaterial analog of a gravitomagnetic monopole". In: *Phys. Rev. D* 110 (10), p. 104035. DOI: 10.1103/PhysRevD.110.104035. URL: https://link.aps.org/doi/10.1103/PhysRevD.110.104035.

Software

- H. Forghani Ramandy (2025). *Event Management api*. Tehran, Iran. Version 11.0.0.1. Web application for managing events, hosts and attendees. Tehran, Iran. URL: https://github.com/hsnfrghn/event-management.
- H. Forghani-Ramandy (2025). *HB Sweets Shop Telegram Bot*. Dubai, UAE. version 3.2.4. Telegram Bot. Tehran, Iran. URL: https://github.com/HB-Sweets/hb-sweets-dubai/.
- LA24 Core Team (2021). *Lavin24 Protect Video Player*. Tehran, Iran. Version 76.3.4. Windows application. Tehran, Iran: Lavin24 Coopration. URL: https://www.lavin24.com/.
- LoyalAxis Team (2022). *LoyalAxis CRM*. Tehran, Iran. Version 3.6.1. WordPress CRM plugin. Tehran, Iran: LoyalAxis. URL: https://la24crm.com/.

Proceedings

- Forghani Ramandy, H. and M. Nouri Zonoz (2021). "Light trajectories in metamaterial analog of Schwarzschild spacetime". *Gravitation & Cosmology. General relativity & Optics*. Proceedings of the National Meeting on Gravitation and Cosmology (Physics Department, Shahib Beheshti University, Jan. 28–29, 2021). Tehran: The physics Society of Iran, pp. 24–27. URL: https://www.psi.ir/farsi.asp?page=ngc99.
- Parvizi, A., H. Forghani Ramandy, and M. Nouri Zonoz (2024). "Photon rings in the metamaterial analog of a gravitomagnetic monopole". *Gravitation & Cosmology. General relativity & Optics*. Proceedings of the National Meeting on Gravitation and Cosmology (Physics Department, Damghan University, Jan. 24–25, 2024). Damghan: The physics Society of Iran, pp. 22–25. URL: https://www.psi.ir/upload/1402/ngc1402/pages/proceedings.pdf.